

# Heart Rate Clocked Analog Audio Synthesizer

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## Abstract

Modular audio synthesizers create the ability to customize the signal paths used to generate audio, allowing for the creation of unique sounds and music. Waveforms are used to control the characteristics of other waveforms. When properly controlled, this makes the distortion created in analog circuits desirable. In this design, the unique personal quality of the music is further enhanced by having the user's heartbeat control the sequencer clock via an electrocardiogram. This clock feeds into a series of interconnected modules that can be rearranged by the user to create custom audio effects. These include a voltage controlled oscillator (VCO) to generate audio waveforms; a voltage controlled filter (VCF) to modify the generated audio; a voltage controlled amplifier (VCA) to modulate the audio amplitude; an attack, decay, sustain, release (ADSR) envelope generator to modify the behavior of the prior modules; a low frequency oscillator (LFO) to provide CV input to other modules; and a sequencer to repeatedly step through sets of inputs. This creates a versatile system for generating various audio effects created from the user's choices and physical state.

## 1 Background

A modular synthesizer is a device that produces sound from generating and altering waveforms, as opposed to mechanical motion and airflow like in traditional musical instruments. The first electronic instruments were developed in the early 20th century. However, development of synthesizers as we know them today took off in the 1960s with the Moog modular synthesizer [1]. Moog invented several of the key modular components such as envelopes, filters, and sequencers that have become standard today.

The key principle of modular synthesizers is control voltage; most modules have some aspect of their function that is controlled by an input voltage. For VCO this is the wave frequency, for VCF this is the filter cutoff, and so on. The control voltage can be a simple DC voltage input, but more often it is varying over time, meaning that the output of some modules can be used as input for others.

The electrocardiogram is also a fairly old technology. The first modern EKG was created in 1904 by

Willem Einthoven in the Netherland and consisted of a piece of silver-coated quartz filament attached to a string in a strong magnetic field. When a current (the heart signal in the EKG) was conducted through this quartz filament, it would move slightly and these movements were recorded. The whole system was massive and required five people to operate and two buckets of saline as the electrodes. EKG technology has developed substantially since then, and our system just uses two small electrode pads and a PCB with signal processing logic.

## 2 High Level Description

Because of the fundamental modularity and customizability of analog synthesizers, there are many different ways to connect our modules together. What is shown in this diagram is how we have chosen to implement it. A VCO usually has several different waveform options that a user can choose from; our VCO currently implements square and triangle waves, with sawtooth waves to come. A sequencer is usually the input to the CV for the VCO. A sequencer switches between designated voltages at a defined clock frequency, and after using connecting this to the VCO CV input, the VCO produces a series of notes at the tempo of the clock frequency. Our unique idea is to use a processed EKG signal for the clock frequency so that the artist's body and feelings are integrated into the musical performance: to create dance music, which is usually 120bpm or higher, the musician's pulse must be 120 or higher, meaning their physical state must match the music they are making.

The output of the VCO then passes through a voltage controlled filter, where the control voltage determines the cutoff frequency for filtering. The control voltage can be controlled either by a knob or another signal input, usually a low-frequency oscillator. The Attack Decay Sustain Release (ADSR) module generates CV for the VCA, creating the envelope for the sound wave. Finally, the sound signal is passed through another amplifier to a speaker output.

## 3 Detailed Description

### 3.1 Electrocardiogram Clock (Kat and Ellie)

We have used the EKG board from lab 5 to obtain the EKG signal.

From the EKG printed circuit board, we further do some signal processing in order to make the heart rhythm into a usable clock signal for the sequencer. First, it goes through a high pass filter to eliminate any slow-varying DC offset. Next, we amplify it with an op amp. The EKG signal starts as just 100mV of noise, with a 200mV peak signal; we bring this level up to 2V. The amplification does not get rid of the noise, but

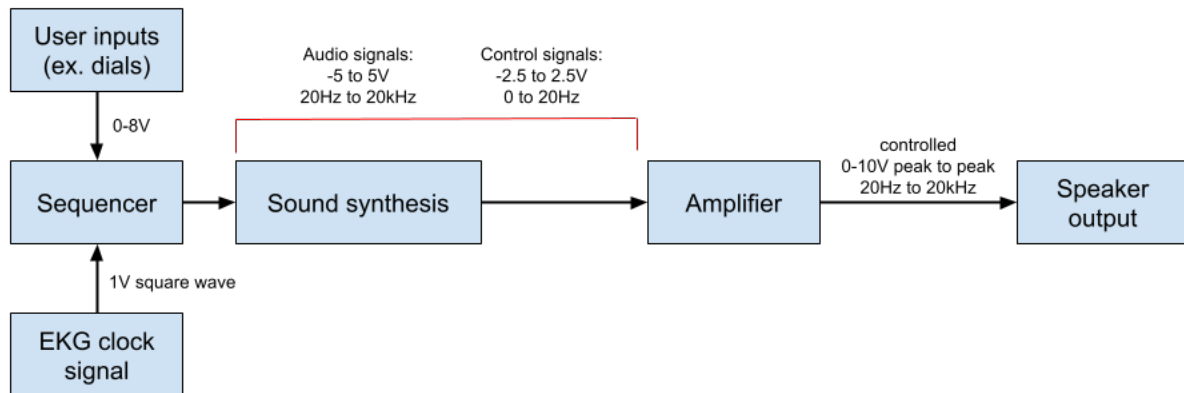


Figure 1: A high level overview of our system

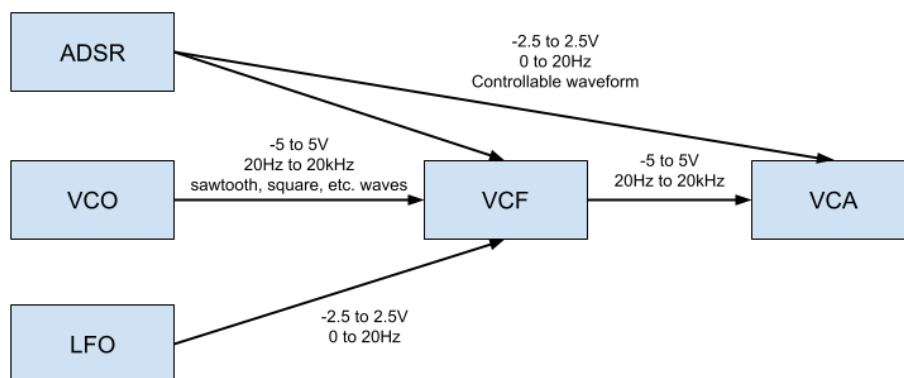


Figure 2: The project's sound synthesis modules

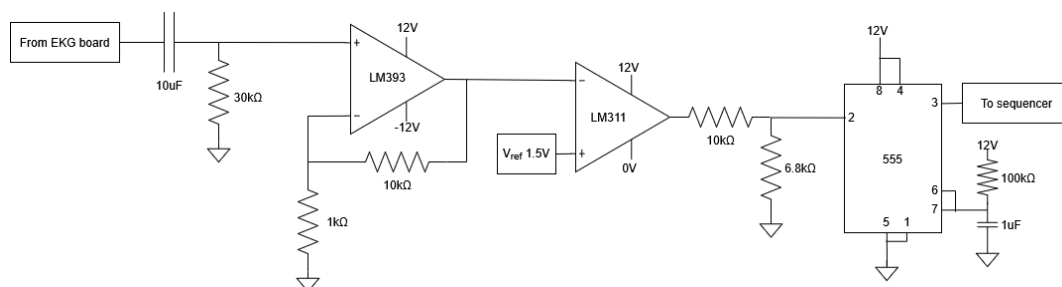


Figure 3: The EKG signal path

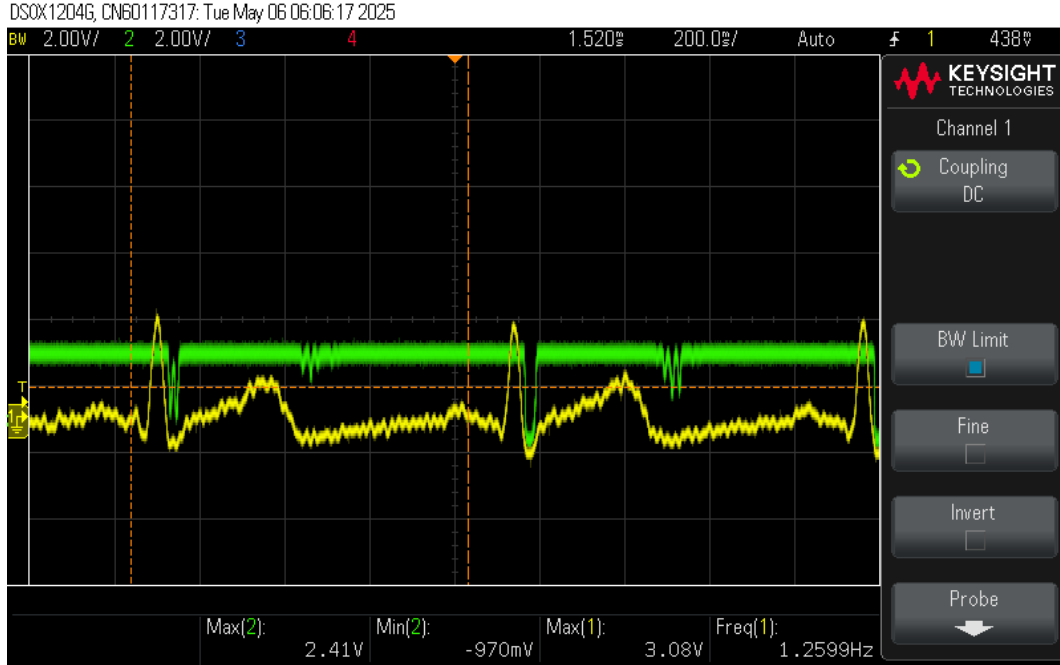


Figure 4: The **EKG signal (yellow)** and **reference voltage (green)**

makes it easier to distinguish the peak for the comparator since the absolute difference between signal and noise is greater.

The amplified signal is fed into a comparator together with a reference signal at approximately 1.5V. The reference voltage is set with a potentiometer to adjust for small variations in the EKG signal from different users. The amplified EKG signal and the reference voltage can be seen in figure 4. At the peak of the EKG signal, it is greater than the reference voltage and the comparator outputs 5V, and otherwise outputs 0V. This square wave with a very small duty cycle (approximately 1%) can now be used as the clock input for the sequencer as it only detects rising edges. The duty cycle is increased using a monostable 555 circuit as part of the sequencer.

### 3.2 Sequencer (George)

The sequencer steps through a series of control voltages corresponding to different audio frequencies in a loop based on the rate of an input clock signal. These control voltages act as an input to the voltage controlled oscillator to generate a repeating melody. The schematic for this element is shown in Figure 5.

The notes, corresponding to control voltages, are selected through a set of  $10k\Omega$  potentiometers with a  $5k\Omega$  series resistor on the high side of the potentiometer so it will output voltages between 0V and 8V to fit the 1V per octave standard.

To select which potentiometer voltage the sequencer will output, the sequencer uses a set of analog

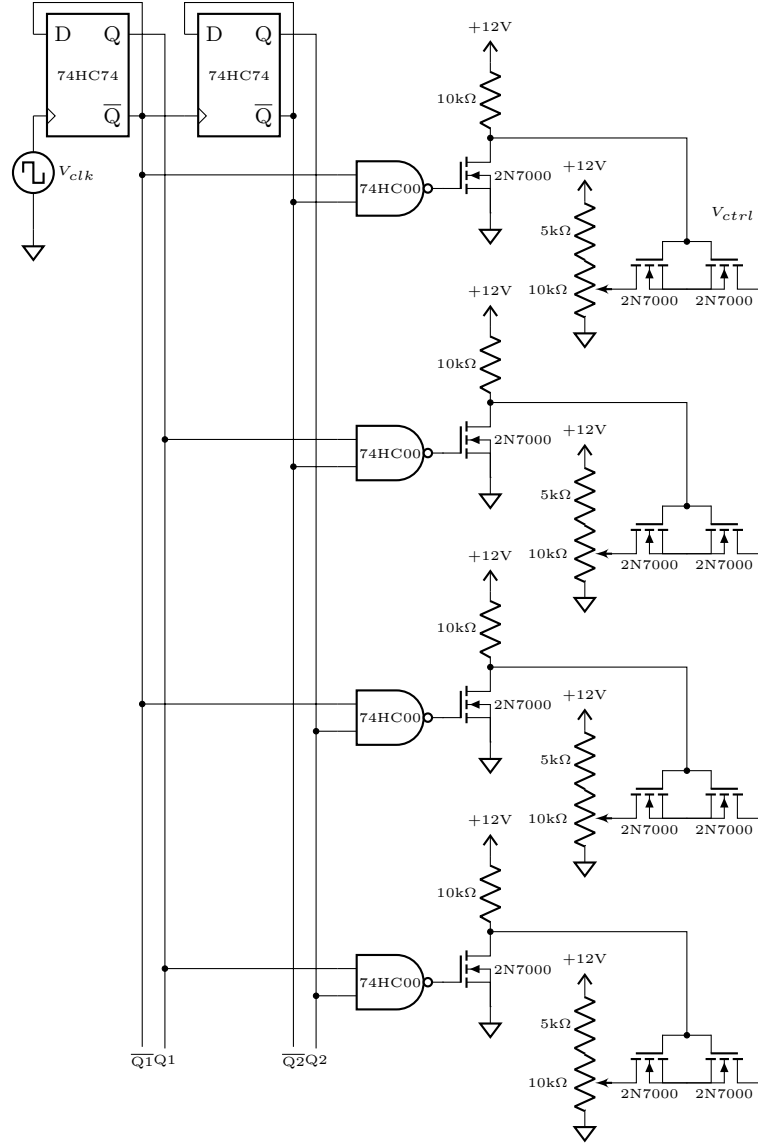


Figure 5: Sequencer Schematic

switches made from back to back pairs of N-Channel MOSFETs to prevent the body diode from conducting so the switch blocks in both directions. Importantly, when using N channel MOSFETs as switches, the gate voltage must be larger than the voltage of the signal being switched. The 2n7000 FETs used in this have a threshold of at least  $0.8V$ , so for a signal that can reach as large as  $8V$ , a gate voltage of at least  $9V$  is required.

To ensure that a sufficiently large gate voltage is provided, even from  $5V$  control signals, a single NMOS transistor inverter is used to drive the gate between  $12V$  and  $0V$ . Notably, this signal is now inverted, with a low signal twinging the switch on, and a high signal turning it off.

To control these analog switches and play notes in order, the sequencer uses 74 series logic chips. The

input clock increments a two bit counter composed of two 74HC74 D flipflops, and the clock generated count is decoded into gate signals for the analog switches by 74HC00 NAND gates to cycle through each note one by one.

The circuit, built on a breadboard, is shown in Figure 6. On the breadboard, but not shown in the schematic is a LM7805 regulator at the top right for generating the 5V power rail for the 74 series logic from the 12V rail the rest of the synthesizer uses. There is also an output buffer on the left to ensure other modules that may draw current from the output control signal will not cause the output voltage to drop.

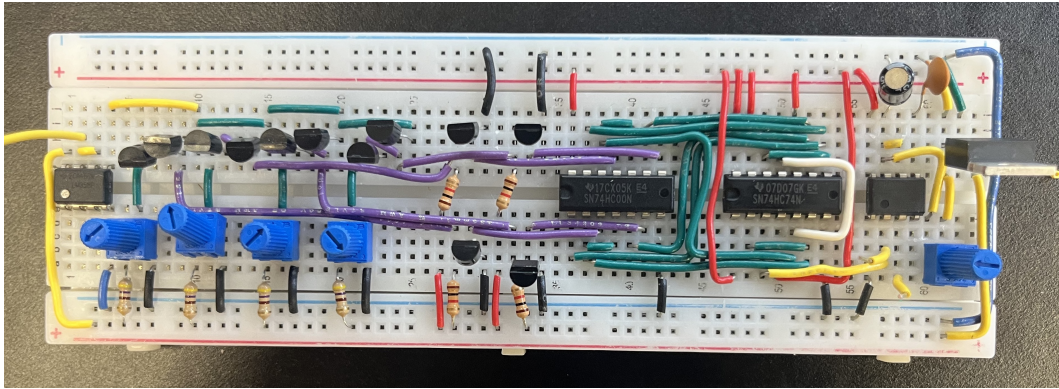


Figure 6: Sequencer Circuit on Breadboard

A final element on the sequencer breadboard that has been absent from Figure 5 is a monostable 555 Timer circuit on the right side of the breadboard (since this circuit is unchanged from the conventional monostable 555 circuit, a detailed discussion of its operation is omitted from this document). This generates an additional output clock with the same frequency of the input clock, but an adjustable high time; this circuit is used to increase the duty cycle of the clock signal coming from the EKG. This gate signal is then used by subsequent modules to change how long a note is played, and the gap between notes.

The resulting output waveforms are shown in Figure 7. The input clock is on channel 1 in yellow, the out gate signal is from channel 2 in green, and the output control voltage is on channel 3 in blue. Each falling edge of the clock switches the gate signal on for a set time, then it falls back to ground. This also changes the control voltage output to the next voltage set by potentiometers creating the stair step output between 8V and ground shown in blue.

### 3.3 Voltage Controlled Oscillator (Ellie)

The voltage controlled oscillator is implemented with an astable 555 timer circuit to produce a square wave. This wave goes between 0V and 12V, but for the following modules, the signal needs to be centered around zero, so there is a high pass filter with  $R=68k$  and  $C=0.1\mu F$  to remove the DC offset. The output of this

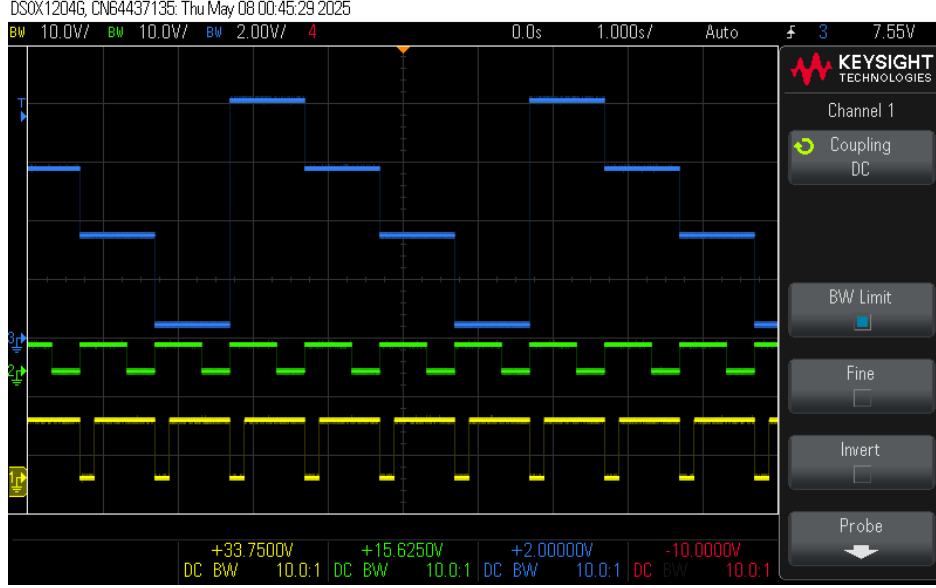


Figure 7: Sequencer Waveform

filter is buffered for use in following modules. The square wave input is also passed to an integrator to create a triangle wave with  $R=10M$  and  $C=0.01\mu F$ . The square wave duty cycle varies with frequency, with decreased on time at higher frequencies, creating an interesting sonic effect.

### 3.4 Attack Decay Sustain Release (ADSR) Envelope Generator (George)

The sound produced by musical instruments, which to a limited extent this synthesizer seeks to mimic, tend to have a specific shape, or envelope, to the amplitude of the notes they create. Using a piano as an example, when a key is first played it produces the loudest sound, which then fades down to a more constant volume until the key is released and the sound fades away entirely.

The magnitude of this volume over time creates an envelope similar to that shown in Figure 9 produced by the ADSR envelope generator used in the synthesizer. The envelope first "attacks" upward to its largest peak, then "decays" down to a chosen "sustain" level until the note ends and the envelope "releases" back to ground. The exact shape of these changes in volume are a matter of preference. In the case of this synthesizer, the attack and decay are linear, while the release is an exponential decay.

The ADSR achieves this shape by conditionally putting charge on and off of an output capacitor through a current source, sink, and a simple resistor, shown in Figure 10.

The attack rate is controlled by a potentiometer as a part of a P-channel JFET current source. The circuit uses a 2N5460 JFET that allows current to flow when  $V_{GS} = 0$ . This creates a voltage drop across the potentiometer increasing  $V_{GS}$  which decreases  $I_{DS}$ . But as the current decreases, so does the drop across

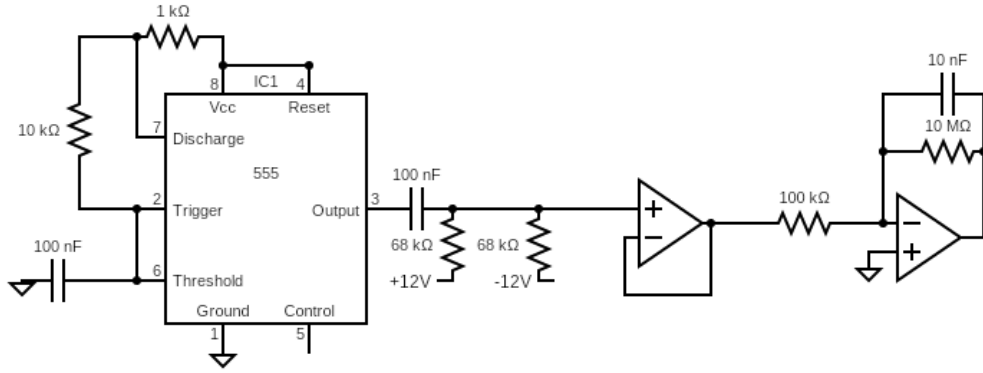


Figure 8: VCO Schematic

the potentiometer, decreasing  $V_{GS}$  and increasing  $I_{DS}$ . All of this is a round about way of illustrating the negative feedback used by this topology, creating a constant current source. The same is also the case on the low side using an N-channel JFET.

The exact current value, dependent on the potentiometer resistance, is rather simple to calculate using the formula given by Equation 1, but for this project a  $10k\Omega$  potentiometer was chosen somewhat arbitrarily as a reasonable value through a trial and error of easily available options. Similarly, the potentiometer resistance for the release stage was chosen to be  $10K\Omega$  since it produced a preferred range of behaviors in the waveform.

$$R = \frac{V_{GS}}{I_{DS}} = \frac{-V_{GS,oss} \left(1 - \sqrt{\frac{I_{DS}}{I_{DSS}}}\right)}{I_{DS}} \quad (1)$$

Connecting the current sources to the capacitor are a series of NFET switches to control when the ADSR envelope should be rising, falling, or simply sustaining its current value. An additional series diode is also placed on each current source to block current from going the wrong direction through the NFET body diodes. Much like the sequencer, these gates require a voltage at least 0.8V higher than the peak of the ADSR envelope.

These control signals are generated by the gate signal produced by the sequencer and two comparators shown in Figure 11. Because the comparator used was an LM393, the output is open collector, meaning it connects the output to ground when low, and gives the output a high impedance when high. Similarly, the gate signal from the sequencer is a 5V signal. For these to drive the gates of the NFET switches they must be amplified, in this case using a simple opamp in an inverting or non-inverting topology depending on what it required by the switch.

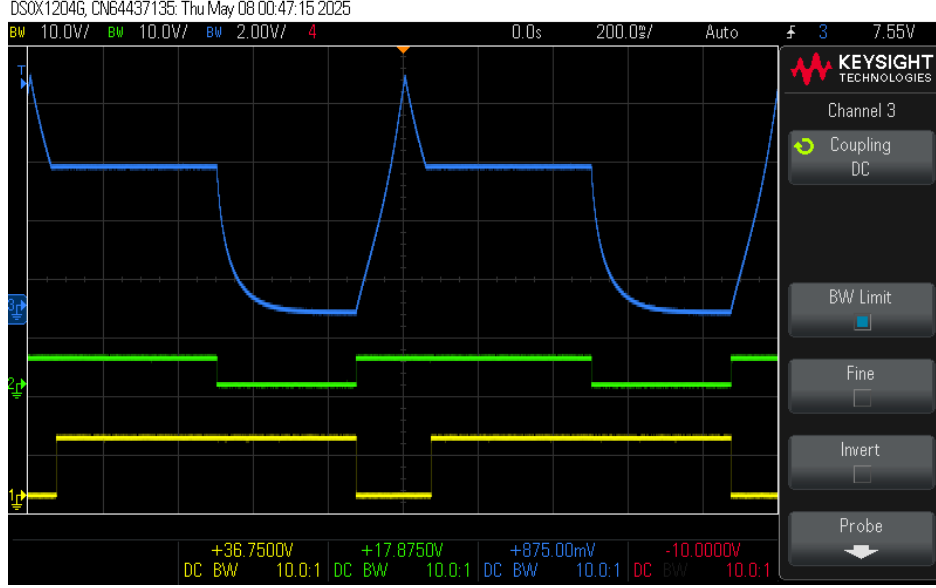


Figure 9: ADSR Waveform

The controls of the envelope generator should only connect the attack current source to the capacitor from when a note starts playing until it reaches the specified attack height. This is done by placing two switches between the current source and capacitor. The first is turned on when the gate signal is high, and thus a note should be played. The second is connected to the comparator circuit producing the signal  $V_a$ . This signal stays high until the output reaches the peak attack height, set by the potentiometer in its feedback network, then it stays low until the output falls back to ground. It achieves this by using the comparator in positive feedback as a Schmidt trigger. By putting these gates in series, the attack current source will only stay on from the start of a note being played until it reaches the specified attack height.

The decay signal works similarly with two series gates requiring two signals to be high to function. The first of these signals is  $V_b$  which opens its gate whenever the output voltage is above the sustain height set by this circuit's potentiometer. This allows the decay current source to be disconnected once the sustain height is reached. However, using just this signal would turn on the current sink before the maximum attack height is reached, so the additional gate only turns on once the attack Schmidt trigger has turned off. This results in the decay current sink to be only connected to the capacitor from when the maximum attack height is reached until the output falls down to its selected sustain level.

During the sustain duration, the voltage does not change since all current paths from the capacitor are disconnected. This means the voltage remains constant until the gate signal falls to ground indicating that the note should stop being played. This closes the release gate allowing the capacitor to discharge through the release potentiometer and decay exponentially to ground.

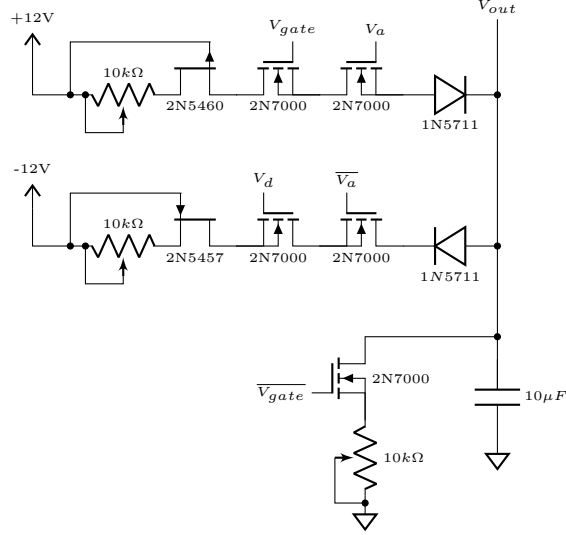


Figure 10: ADSR Capacitor Charging/Discharging Schematic

The resulting waveform from this process is shown in Figure 9 where the output control voltage is in blue, the input gate signal is in green, and the yellow signal represents the clock that is sent to the sequencer. The physical circuit is shown in Figure 12 on a breadboard. Not shown in the schematics but on the breadboard is an opamp output buffer on the left, and an amplifier and inverter for the 5V gate signal that is the third IC from the right.

### 3.5 Low Frequency Oscillator (George)

It may be desirable for further audio frequency modulators to change their behavior periodically. This can be done within a single note to add vibrato, or at a lower frequency to changing sounds across iterations of the melody. This can be done with a low frequency oscillator (LFO) that creates waveforms at frequency from about  $20Hz$  to  $0.1Hz$ . It is unfortunately quite difficult to produce low frequency sine waves. Instead, the LFO produces pulse, triangular, and trapezoidal wave that can be selected between by the user.

The LFO first produces a triangle wave that in turn gets modified into a pulse and trapezoid wave. Much like the ADSR, the linear rise and fall of the triangle wave is produced through constant current sources that are switched between using a Schmidt trigger. The Schmidt trigger oscillates from  $12V$  to  $-12V$  and triggers when the capacitor voltage reaches  $2V$  and  $-2V$ , creating a triangle wave between  $-2V$  and  $2V$ .

Unlike the ADSR (which uses JFET current sources), the LFO uses current sources formed from an opamp, a resistor, and a BJT as shown in Figure 13. A blocking diode is also used to enforce the direction of the constant current sources. These sources each use one half of a potentiometer as their series resistor with a wiper voltage  $V_{ctrl}$  that oscillates between  $+12V$  and  $-12V$ . The significance of these properties will

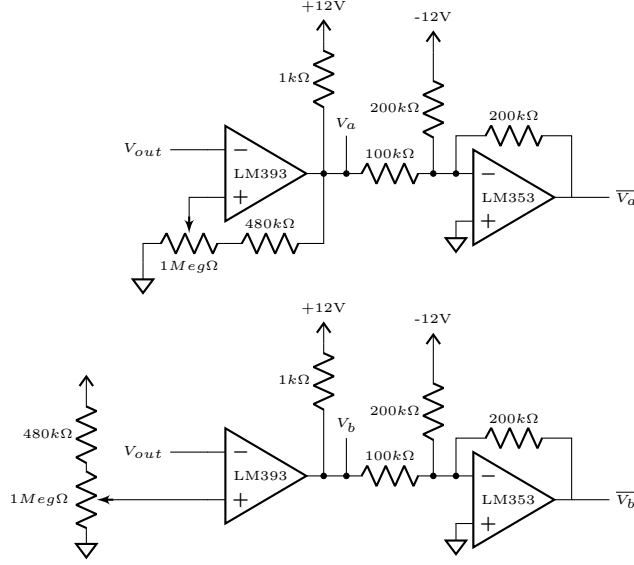


Figure 11: ADSR Comparator and Schmidt Trigger Schematic

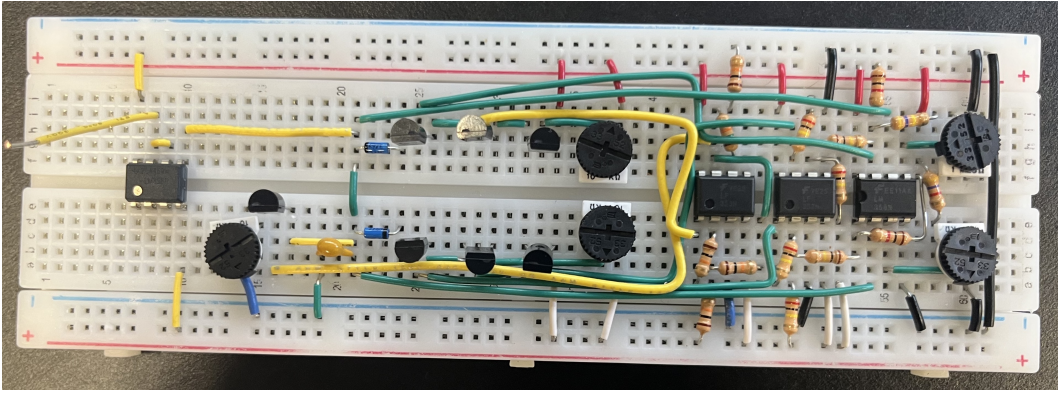


Figure 12: ADSR Circuit on Breadboard

be investigated later on.

Assuming  $V_{cmp} = -12$  the capacitor should discharge at a constant current. This is enforced through the negative feedback of the op-amp. The op-amp drives the NPN BJT such that the potentiometer voltage seen by the inverting output of the op-amp is the same as the voltage on the non-inverting in out. This keeps a constant voltage from one side of the potentiometer to the wiper is  $\alpha V_{pot} = V_{amp,+} + 12V$  and thus a constant current  $I = \frac{V_{amp,+} + 12V}{\alpha R_{pot}}$ . Importantly, this current has a linear relationship to the non-inverting voltage of the opamp and an inverse relationship with potentiometer resistance.

The non-inverting op-amp voltage is set by using a second op-amp circuit to subtract a frequency control voltage  $V_{freq}$  to  $V_{ctrl}$  for charging and add them when discharging. This results in a charge and discharge currents that are proportional to  $V_{freq}$ . This makes a charge current  $I_{charge} = \frac{V_{freq}}{\alpha R_{pot}}$  and a discharge



To get a pulse wave, the output of the Schmidt trigger can be used seen in green in figure 14. And finally shown in yellow, the trapezoid waveform is produced by amplifying the triangle wave with an op-amp such that the signal clips at  $+12V$  and  $-12V$ . Each of these signals is put through their own amplifier to normalize their amplitudes (not shown in the schematic). Then a series of switches is used to select between these waveforms which get put through one final summing op-amp to let the user change the magnitude and DC offset of the output. This final circuit is shown built on a breadboard in figure 15.

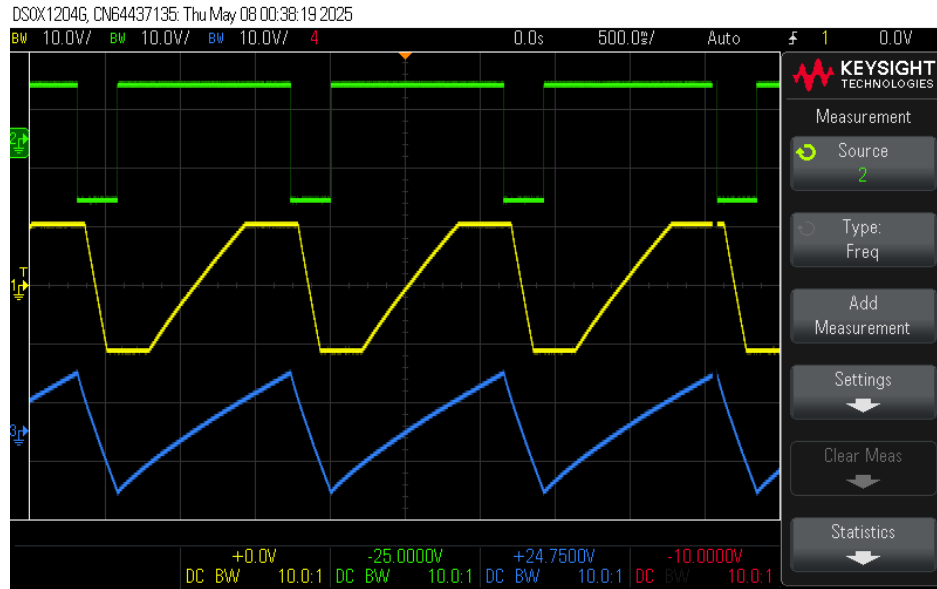


Figure 14: LFO Waveform

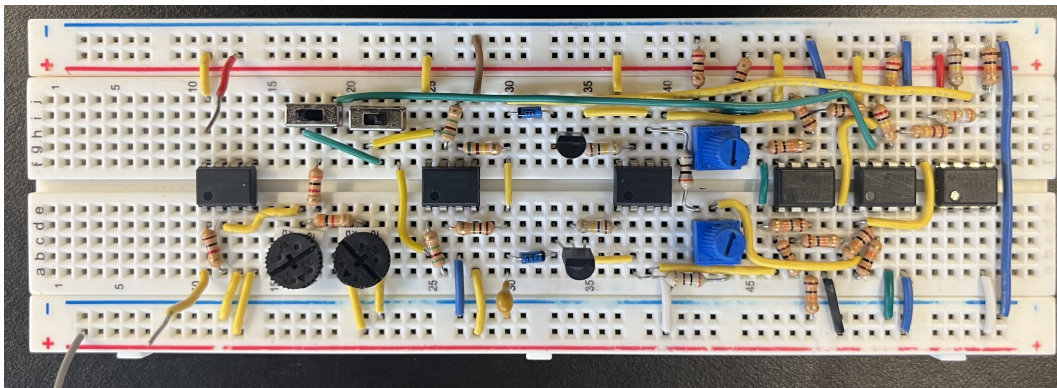


Figure 15: LFO Circuit on Breadboard

### 3.6 Voltage Controlled Filter (Ellie)

The voltage controlled filter is a filter whose cutoff can be varied by a control voltage. It takes in the sound wave and removes the higher frequency components, which, for instance, turns a square wave into a wave with more rounded corners. In the design we are using, the control voltage can be modified both by a knob (connected to a potentiometer) and by an input signal such as an LFO. Using an LFO for the control voltage allows the filter cutoff to vary over time without continual input adjustment. The challenge of analog filter design is that most types of filters depend on resistance and/or capacitance, and it is difficult to use a control voltage to modify either of these characteristics. However, two main solutions have been found: transistor ladders and diode ladders. The heart of this design is a diode ladder. A diode has a non-linear IV curve whose slope is always non-negative, so it can be considered as having increasing resistance with

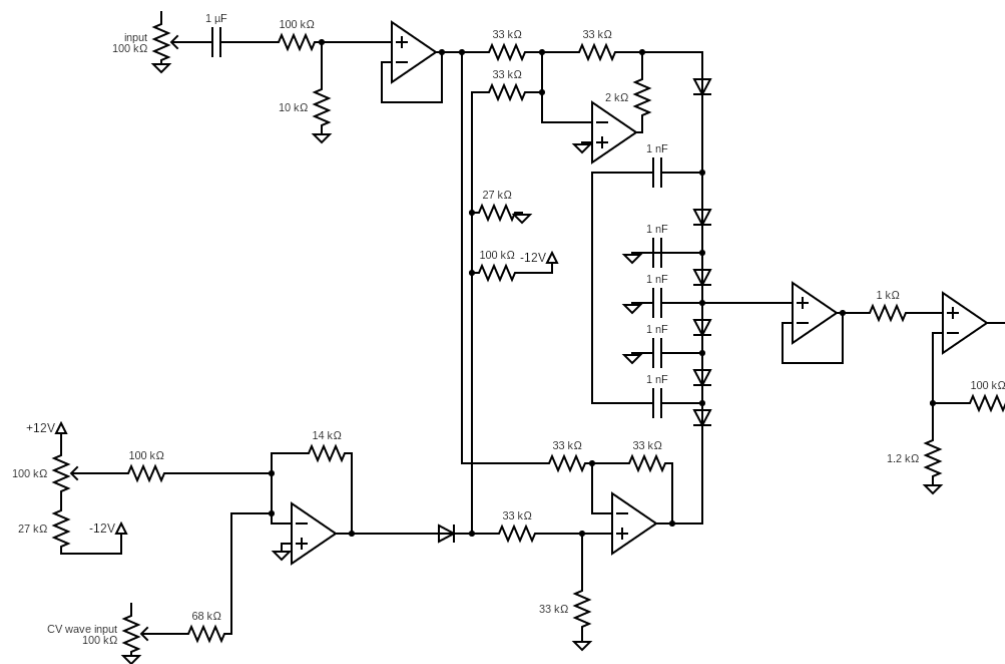
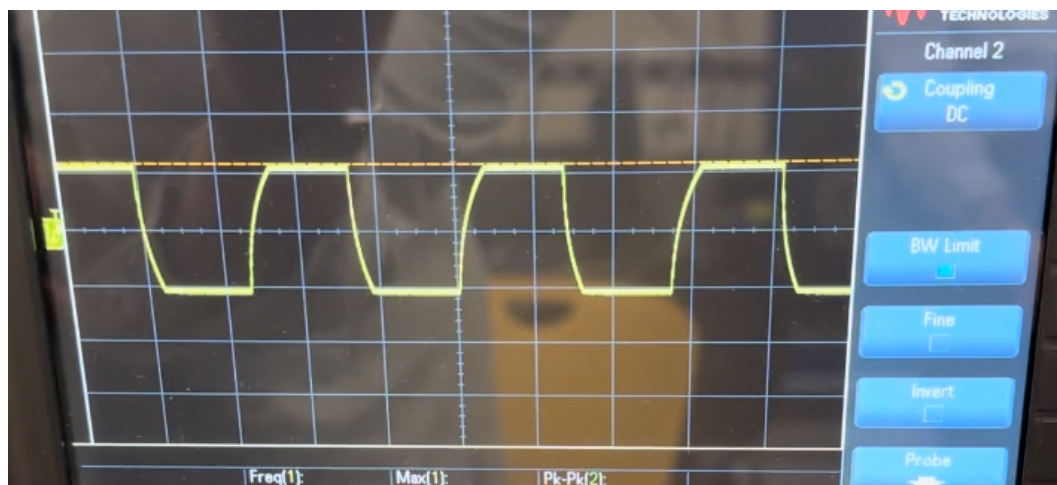


Figure 16: VCF Schematic



increasing voltage (in a non-linear manner), and thus the control voltage can control the effective resistance of the diode which yields a different cutoff frequency. The input stage AC couples and then attenuates the signal so as to not overload the diodes with too much current and then has two op-amps circuits to raise and lower the DC voltage to connect to either end of the diode ladder respectively. The output stage then re-amplifies and buffers the filtered signal. The pictured waveform is a lightly filtered square wave; the filtering is apparent from the rounded edges of the wave, indicating that some of the high-frequency components have been filtered out. This design is substantially drawn from the Erica DIY Synths VCF design with some modifications to the input and output stages as well as some resistor/capacitor values.

### 3.7 Voltage Controlled Amplifier (Kat)

The voltage controlled amplifier (VCA) takes in the sound wave, and modifies the amplitude based on two control voltage signals. The control voltages (CVs) are a 0-5V signals and can come from any of the previous modules, allowing us to modify this output in many different ways. They are controlled with two potentiometers to change how much each affects the amplitude.

At a high level, the VCA converts the CVs into a current signal using a BJT and op amp (TL072) current source, and feeds this current into the  $I_{abc}$  gain pin of the operational transconductance amplifier (OTA). The sound signal from previous modules goes into the inverting input, and the non-inverting is grounded. The result is that the sound is amplified proportionally to the CVs. The OTA output is also a current signal and we convert that into a voltage using an TL702 op amp. Lastly, the output of the VCA is passed to our power stage to amplify it again for the speaker output.

### 3.8 Power Amplifier (Kat)

The last stage of our modular synthesizer is a power amplifier which amplifies the sound to be played on a speaker. We use a class B push-pull amplifier with two BJTs (an 2n304 and 2N906). There is a feedback loop going back to a TL072 op amp to prevent crossover distortion when the signal passes ground. The volume is adjusted with the 100K potentiometer, and the component values were chosen based on the maximum power rating of our speaker (0.5W). Since it's a  $32\Omega$  speaker, we designed the circuit to provide at most 3.6V so that the current is 0.113A, and the power stays below 0.4W.

## 4 Potential Improvements

One big next step for the project would be making the EKG more robust to user heart signal differences. At the moment, switching between users requires tuning the comparator reference voltage to adjust for

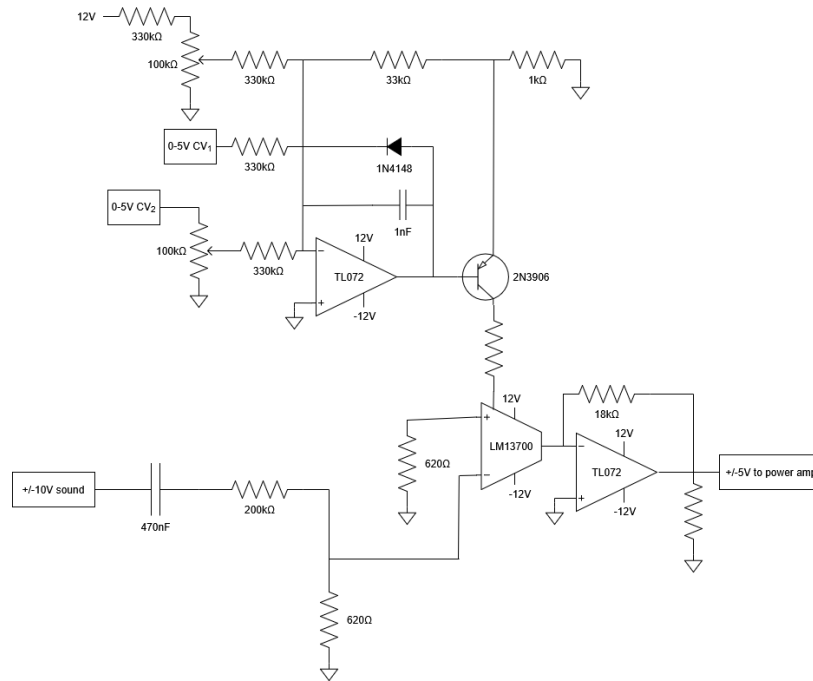


Figure 17: The voltage controlled amplifier

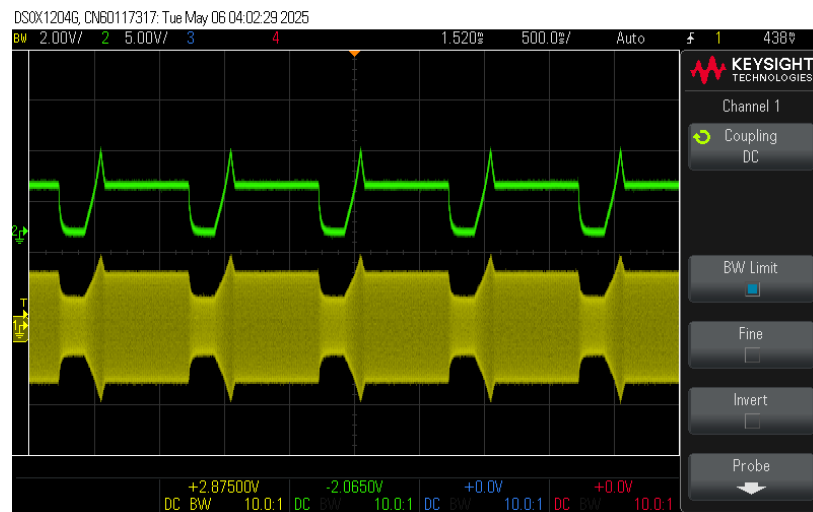


Figure 18: The audio output signal with modified amplitude (yellow) and the ADSR control voltage (green)

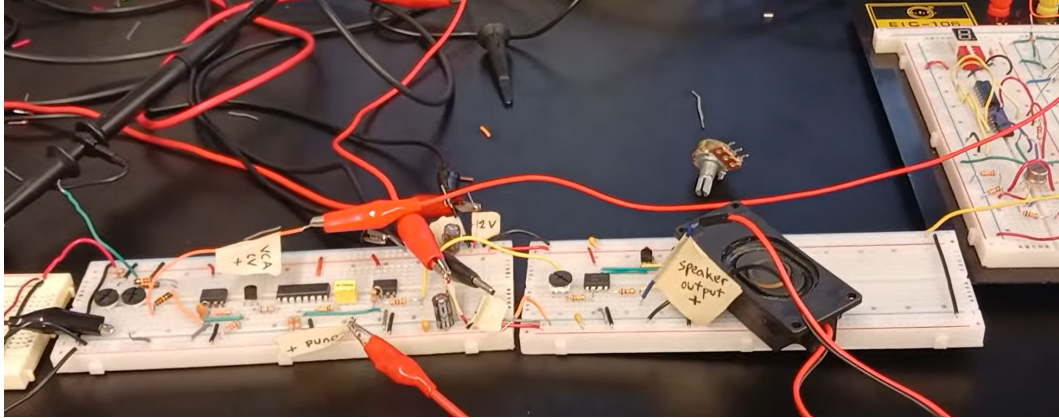


Figure 19: The VCA and power amplifier on a breadboard

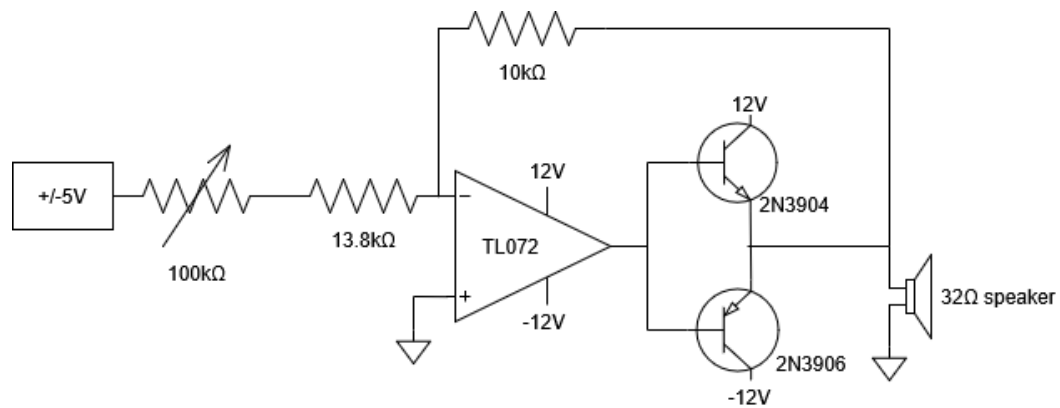


Figure 20: The schematic of the power amplifier for speaker output

differences in signal strength. Some alternatives could be using a differentiator circuit to detect the spike in the EKG graph, or some digital signal processing. Another improvement for the project would be to add a quantizer to “snap” sequencer control voltages to musical notes, which would make tuning the synthesizer easier. Another change we could make to the existing modules would be to add more waveform shapes to the VCO. Finally, building these modules on a PCB instead of breadboards would be a useful next step for reducing signal noise and making the synthesizer easier to use.

## 5 Conclusion

The project successfully demonstrated the synthesizer ability to generate a melody modulated with the outlines methods, though the musical quality of that melody is debatable. The project successfully demonstrates the team's ability to design and build tunable oscillators and modulators at audio frequencies, and generate control waveforms at low frequency. While the design of every board has room for optimization, and any long term use would require the circuit be transferred from breadboards to PCBs, the fundamental circuit designs were quite reliable. Overall the project was quite successful.

In terms of personal value, the project acted as a valuable source experience. Our members' experience with project based classes spanned the range from those who have done many CIM classes requiring group design work, to a member for which this was their first CIM. This range of experience was valuable for the project's success, but more importantly the learning of group members. The organization of the engineering aspects of the project went rather smoothly assisted by the module based nature of the project allowing work to be divided easily, avoiding many of the scheduling issues common in projects. However clear communication at the outset on how each component should behave was vital to the project's ability to work when put all together. Additionally, the CIM nature of the class required more extensive documentation which was useful practice. With so many components needing to be put together between different people documentation was vital for keeping track of how things worked and were setup over the course of the project. Overall while many of the formal aspects of the project organization and documentation were annoying to deal with the experience had value.

## Acknowledgment

We would like to thank Mike, Tiffany, Soojung, and Alec for their help on this project.

## References

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